

Learning goals: Plots in R. Wherever necessary please code in the actual PDF, do not use intrinsic functions to sample distributions in this lab. This lab uses the data in SOCR-HeightWeight.csv

EXERCISES

- (1) Plot the point (2,4) with square point character type and magenta color.
- (2) Create a sequence of function values corresponding to $\sin(x)$ and $\cos(x)$ function from $-\pi$ to π and plot the two functions on the same graph with appropriate titles, axes labels, blue color for the sine curve, red color for the cosine curve, star point character type for sine, cross type point character for cosine overlaid by lines joining the points.
- (3) Reproduce the bar graph type of plot in Fig. 4.2.1 in the Biostatistics book by Daniel using the appropriate settings for the appearance.
- (4) Make a 2x3 grid of 6 graphs with the following specifications and their respective titles (keep the title short!)
 - (i) x vs $\cos(x)$ with red color and lines
 - (ii) x vs $(x^2/3) + 4.2$ with violet color, points and lines, linewidth 2 and linetype 1.
 - (iii) Histogram plot of binomial distribution for 12 trials and $p=0.3$
 - (iv) Histogram plot of binomial distribution for 12 trials and $p=0.8$
 - (v) Histogram plot using `type='h'` option in `plot()` function for x sequence of 1 to 10 with 0.5 spacing and y function $50x/(x+2)$ with colors blue and orange alternating between the points.
 - (vi) x vs $\log(x)$ with orange color and 'step' linetype.
- (5) Write a script to recreate the plot in the slide with the plot title 'This is a graph'.
- (6) Plot a histogram representation of hypergeometric distribution with $N=500$, $K=50$ and $n=30$
- (7) Write few lines of script to explore what happens to a hypergeometric distribution when n is increased and gets closer to N . Show that it approaches the binomial distribution by plotting histograms of both hypergeometric and binomial on the same plot. Use a 3x3 grid of 9 graphs to show the convergence of hypergeometric to binomial distribution.
- (8) On the same plot, draw 3 Poisson distributions with λ values of 3,20,45 (Code the probability distribution function).
- (9) Load the csv file for heights and weights of 25000 people and do the following:
 - (i) Plot a histogram of the height variable and determine its mean and standard deviation
 - (ii) Plot a histogram of the weight variable and determine its mean and standard deviation
 - (iii) Draw a Gaussian curve (recall the Gaussian PDF) with the above calculated mean and standard deviation for both height and weight variables as Z vs $P(Z)$ (i.e. Z -transformed). Plot using either `plot()` function or `curve()` function.
 - (iv) What happens when you decrease the size of the bins in the histogram plots? Make a 1x3 grid of 3 plots that show the trend for decreasing bin sizes.
- (10) Plot the PDF and CPDF for the uniform distribution $U(1,2)$. Find a way to shade the region under the PDF up to $x = 1.5$.
- (11) Plot the PDF and CPDF for the exponential distribution with $\lambda = 10$. Shade the region under the PDF up to $x = 2.8$.
- (12) Plot the PDF and CPDF for the Gamma distribution with $\alpha = 5$ and $\theta = 3$.
- (13) Plot the PDF and CPDF for the Chi-square distribution for 20 degrees of freedom. Shade the region under the PDF up to $x = 1.0$.